ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
Department of Mechanical and Production Engineering



MIC LAB (ME 4408)
Project Report

Gesture Controlled Wheelchair

Group 02

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Abstract

This report presents the design, construction and testing of a Gesture Controlled Wheelchair, an assistive mobility device that lets a person with limited mobility steer a wheelchair by tilting the hand instead of operating a conventional joystick. The system has two units. The hand (transmitter) unit is a glove carrying an MPU-6050 accelerometer–gyroscope, an Arduino Nano and an HC-05 Bluetooth module; it measures the orientation of the hand, converts it into a motion command and transmits it wirelessly. The wheelchair (receiver) unit uses a second HC-05 module, an Arduino Uno and an L298N dual H-bridge driver to run the DC gear motors so that the chair moves forward, backward, left or right, or stops. A proof-of-concept chair frame was modelled in CAD and fabricated from PVC pipe. Tests showed that the gestures were interpreted reliably and that the chair responded smoothly at a constant, manageable speed in all four directions. The prototype was built for an estimated component cost of about Tk 2,815. Speed control, camera-based autonomous navigation, collision avoidance and an assistive robotic arm are identified as future improvements.

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1 Introduction

1.1 Background

Mobility is central to independence and quality of life. Many people with mobility impairments depend on wheelchairs, and powered wheelchairs are usually driven with a hand-operated joystick. For users with weak grip, tremor or limited fine-motor control, a joystick can be tiring, imprecise or impossible to use.

The Gesture Controlled Wheelchair project offers an assistive technology solution that lets such users move around using intuitive hand gestures. By using gesture recognition, the system allows a wheelchair to be controlled with precision and ease, giving a more natural alternative to traditional joystick control [1].

1.2 Problem Statement

Conventional powered wheelchairs require continuous, fine manipulation of a joystick. A low-cost control method is needed that

- recognises simple, natural hand movements;
- converts them reliably into wheelchair motion commands;
- works wirelessly so that the user is not tethered to the chair; and
- can be built from affordable, readily available components.

1.3 Scope of the Report

This report describes the objectives of the project (Chapter 2), a short review of related work (Chapter 3), the selection of the project (Chapter 4), the system design and components (Chapter 5), the working principle and control algorithm (Chapter 6), fabrication and assembly (Chapter 7), cost analysis (Chapter 8), results and discussion (Chapter 9), and finally the conclusion and future scope (Chapter 10).

2 Objectives

The main objective of the Gesture Controlled Wheelchair project is to develop an intuitive and accessible assistive technology that enables people with mobility impairments to control their wheelchairs using hand gestures [2]. The specific objectives are:

1. to design a wearable hand unit that senses hand orientation with an MPU-6050 inertial sensor;
2. to transmit gesture commands wirelessly over Bluetooth to the wheelchair;
3. to drive the wheelchair forward, backward, left and right, and to stop it, according to the received command;
4. to build and test a working proof-of-concept prototype at low cost.

Additional key objectives of the project include:

1. enhanced independence for the user;
2. enhanced mobility through gesture-based control;
3. increased convenience and comfort; and
4. improved quality of life.

Figure 2.1 summarises the wider design goals of the system.



Figure 2.1. Objectives of the gesture controlled wheelchair.

3 Literature Review

Gesture-controlled vehicles have been widely studied as a human–machine interface for robots and assistive devices. Gomathy [1] described a gesture controlled robot in which an accelerometer worn on the hand detects tilt, a microcontroller encodes the tilt into a direction command, and a wireless link carries the command to a motor driver on the robot. The same architecture is suitable for a wheelchair because the four basic manoeuvres of a differential-drive vehicle (forward, backward, left and right) map naturally onto four directions of hand tilt.

Anilkumar [2] applied this idea to a hand gesture controlled wheelchair, showing that a wearable inertial sensor and a low-cost microcontroller can replace the joystick for users who find a joystick difficult to operate.

The present work follows the same principle. It uses the six-axis MPU-6050 inertial measurement unit [3], Arduino microcontrollers [6], HC-05 Bluetooth modules for the wireless link [5] and an L298N dual H-bridge motor driver [4], all of which are inexpensive and widely available.

4 Project Selection

Before the final project was chosen, the group drew up preliminary component lists and cost estimates for five candidate microcontroller-based projects. Table 4.1 compares the estimated cost of each.

Table 4.1. Candidate projects considered and their estimated component cost.

No.	Candidate project	Estimated cost (Tk)
1	Gesture controlled wheelchair (selected)	2,815
2	Fire-fighter robot	2,282
3	Soil-moisture based automatic plant watering	1,935
4	Home security system	5,517
5	Solar tracking system	3,980

The gesture controlled wheelchair was selected because it

- addresses a real social need in assistive mobility;
- combines sensing, wireless communication, embedded programming and motor control in one system, which matches the learning goals of the course;
- involves both mechanical design (the chair frame) and electronics, which suits a mechanical engineering laboratory; and
- can be built within a moderate budget of under Tk 3,000.

5 System Design

5.1 Overall Architecture

The system has two independent units that communicate over Bluetooth (Figure 5.1).

- **Hand (transmitter) unit:** an MPU-6050 sensor, an Arduino Nano and an HC-05 Bluetooth module (master) mounted on a glove and powered by a 9 V battery.
- **Wheelchair (receiver) unit:** an HC-05 Bluetooth module (slave), an Arduino Uno and an L298N motor driver that drives the left and right DC gear motors, powered by a battery pack through a switch.

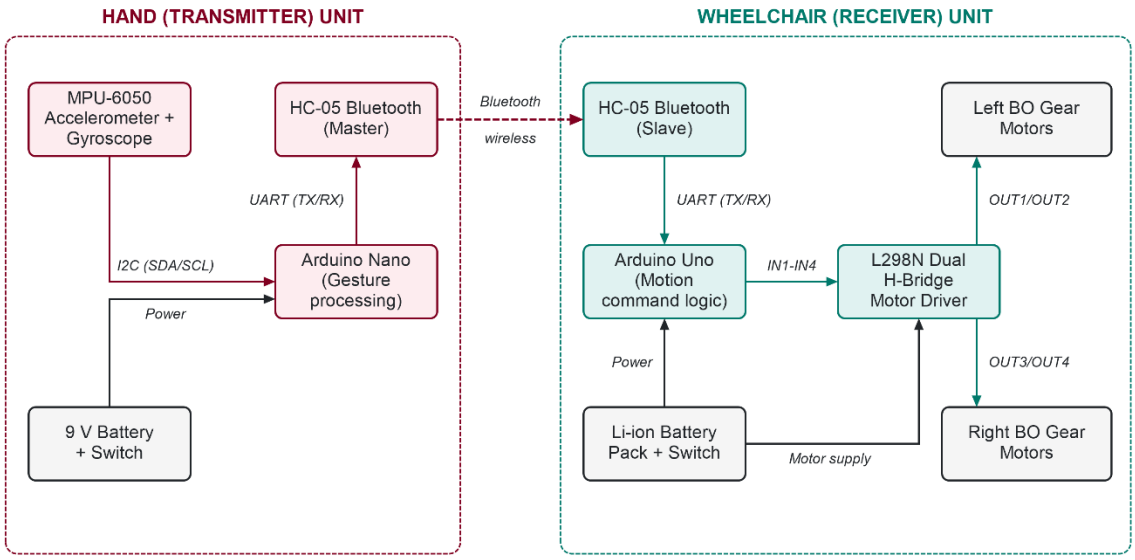


Figure 5.1. Block diagram of the gesture controlled wheelchair system.

5.2 Mechanical Design

The wheelchair frame was first modelled in CAD to fix the overall proportions: seat, backrest, armrests, footrest and four wheel mounts. The frame was then fabricated from PVC pipe and fittings, with board panels for the seat, backrest and footrest (Figure 5.2). The space under the seat is used as an electronics bay that holds the controller, motor driver and batteries. PVC was chosen because it is light, cheap and easy to cut and join, which suits a proof-of-concept prototype.

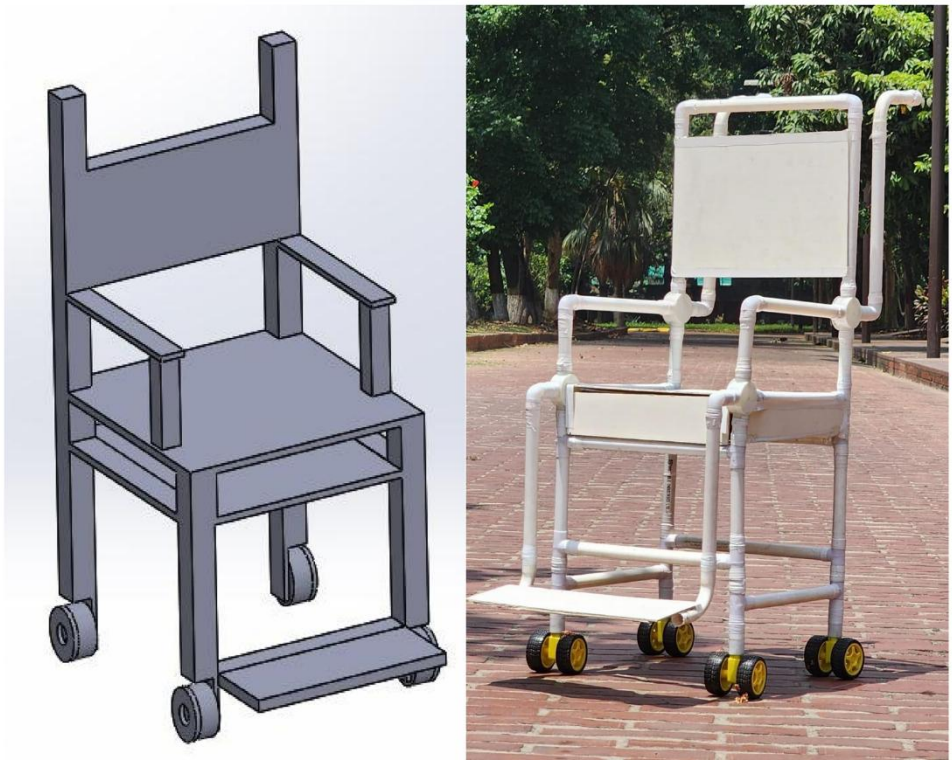


Figure 5.2. CAD model (left) and fabricated model (right) of the gesture controlled wheelchair.

5.3 Electronic Components

Table 5.1 lists the main electronic components and their roles in the system.

Table 5.1. Main electronic components and their functions.

Component	Key characteristics	Function in the system
MPU-6050	3-axis accelerometer and 3-axis gyroscope; I2C interface [3]	Senses the tilt (orientation) of the user's hand
Arduino Nano	ATmega328P, 16 MHz, compact board [6]	Reads the sensor, classifies the gesture and sends the command
HC-05 Bluetooth (2 pcs)	Serial-port Bluetooth module, configurable as master or slave [5]	Wireless link between the hand unit and the wheelchair
Arduino Uno R3	ATmega328P, 16 MHz, 14 digital I/O pins [6]	Decodes the received command and sets the motor-driver inputs
L298N motor driver	Dual full H-bridge driver [4]	Controls the direction of the left and right motors
BO DC gear motors	Low-cost plastic-geared DC motors	Drive the wheels of the chair
Batteries and switches	9 V battery (hand unit); battery pack in holder (chair)	Power supply with manual on/off isolation

5.4 Wheelchair Circuit

The wheelchair circuit was assembled carefully to ensure correct operation and dependability (Figure 5.3). The Arduino Uno, the main control unit, is fixed firmly to the wheelchair frame. A reliable connection between the Arduino Uno and the Bluetooth module allows the wheelchair to be controlled remotely. The motors and the Arduino Uno are wired to the motor driver, which is essential for motor control. Battery switches are included in the circuit to control the power supply and to allow the wheelchair to be operated safely.

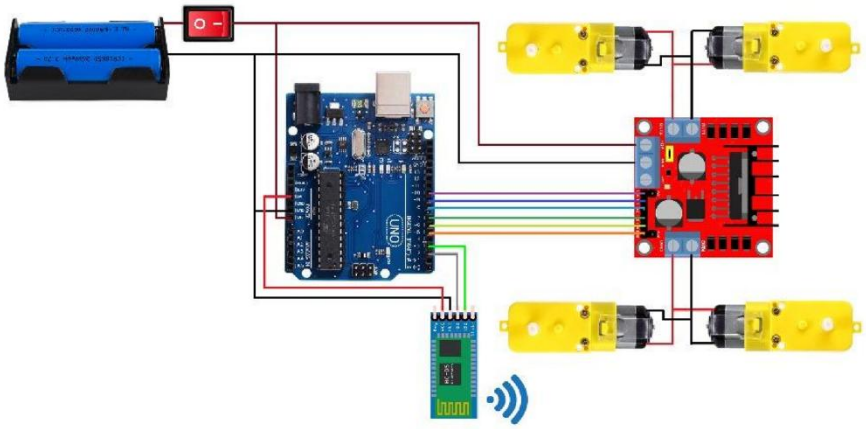


Figure 5.3. Schematic diagram of the wheelchair (receiver) circuit.

5.5 Hand Gesture Circuit

To save space and keep the wiring short, the Arduino Nano, battery, MPU-6050 accelerometer–gyroscope and Bluetooth module of the hand unit are arranged neatly on a breadboard (Figure 5.4). Each component is placed to minimise interference and to give the best possible signal transmission and reception. The wiring connections are soldered or firmly fastened to avoid loose contacts and signal degradation, and suitable insulation protects against electrical hazards and short circuits [1], [2].

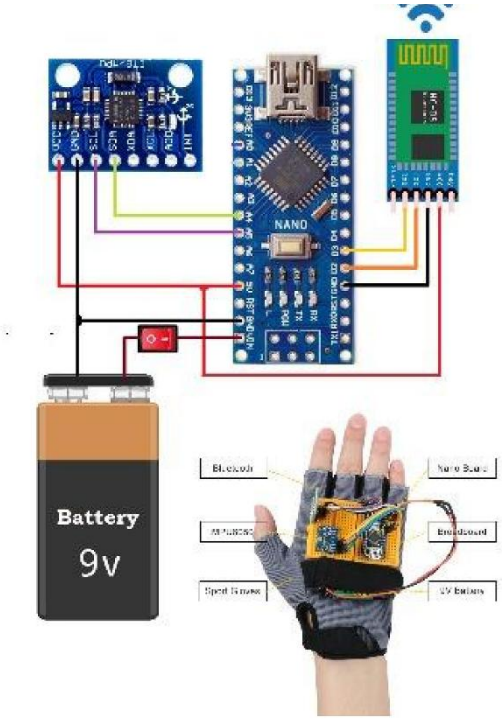


Figure 5.4. Schematic diagram of the hand (transmitter) circuit and its mounting on the glove.

6 Working Principle and Methodology

6.1 Control Algorithm

The Arduino code reads and interprets the gesture data from the MPU-6050, and Bluetooth communication carries the signal between the hand gesture circuit and the wheelchair [1], [2]. The complete process follows the seven steps shown in Figure 6.1, and steps 2–7 repeat continuously while the system is switched on.

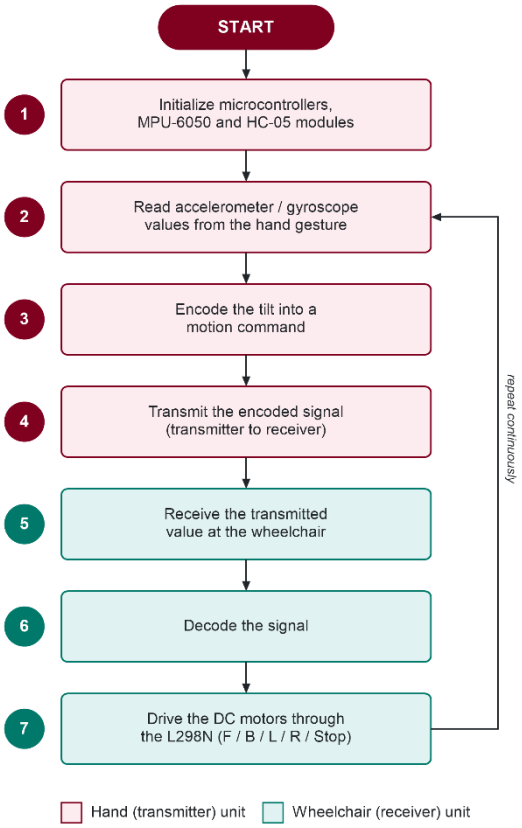


Figure 6.1. Flowchart of the gesture controlled wheelchair.

6.2 Gesture Recognition

The MPU-6050 measures acceleration and angular rate along three axes. When the hand is tilted, the direction of gravity relative to the sensor changes, so the tilt of the hand (pitch and roll) can be found from the accelerometer readings. The Arduino Nano compares the tilt with preset thresholds and assigns one of five commands, as listed in Table 6.1. The thresholds were tuned experimentally so that small, unintended hand movements do not move the chair.

Table 6.1. Mapping of hand gestures to wheelchair motion.

Hand gesture	Wheelchair response
Hand tilted forward	Moves forward
Hand tilted backward	Moves backward
Hand tilted to the left	Turns left
Hand tilted to the right	Turns right
Hand held level (neutral)	Stops

6.3 Wireless Communication

The two HC-05 modules are paired, the hand unit's module acting as master and the chair's module as slave. Each module is connected to the serial (UART) pins of its Arduino, so a command written to the serial port of the Nano appears at the serial port of the Uno. Each gesture is encoded as a single command character, which keeps the data short and the response fast.

6.4 Motor Control

The Arduino Uno decodes the received character and sets the four direction inputs (IN1–IN4) of the L298N. The left-side motors are connected to one H-bridge channel and the right-side motors to the other. Turning is achieved by differential drive, i.e. by running the two sides in opposite directions. Table 6.2 gives the driver logic used for each motion.

Table 6.2. L298N input logic for each motion (1 = HIGH, 0 = LOW).

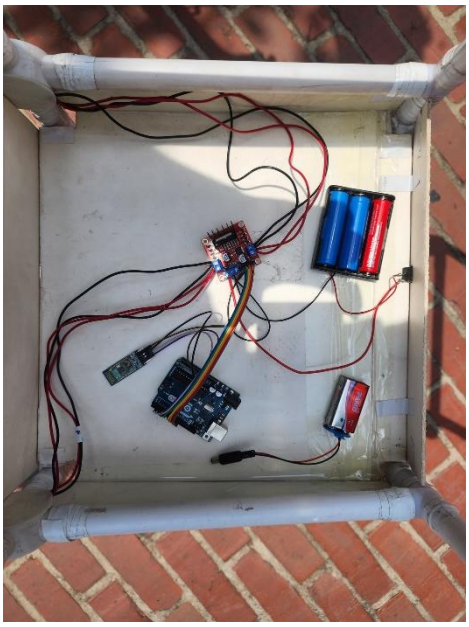
Motion	IN1	IN2	IN3	IN4	Left motors	Right motors
Forward	1	0	1	0	Forward	Forward
Backward	0	1	0	1	Backward	Backward
Left turn	0	1	1	0	Backward	Forward
Right turn	1	0	0	1	Forward	Backward
Stop	0	0	0	0	Stopped	Stopped

The straight-line and turning speeds are kept constant at a moderate, manageable level for safety and ease of control.

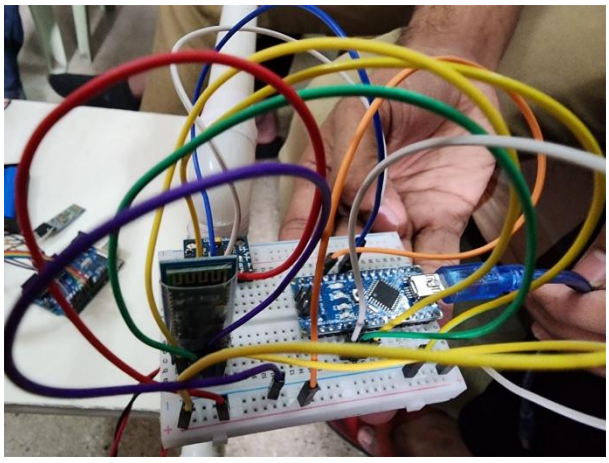
7 Fabrication and Assembly

Fabrication was carried out in the following stages:

1. **Frame construction:** PVC pipes were cut to the lengths taken from the CAD model and joined with elbows, tees and cross fittings to form the seat, backrest, armrests, push handle and legs. Board panels were fixed for the seat, backrest and footrest.
2. **Drive installation:** the DC gear motors and wheels were mounted at the bottom of the legs.
3. **Chair electronics:** the Arduino Uno, L298N driver, HC-05 module, battery pack and switches were fixed inside the bay under the seat and wired as shown in Figure 5.3 (Figure 7.1a).
4. **Hand unit:** the Arduino Nano, MPU-6050, HC-05 module and 9 V battery were wired on a breadboard (Figure 7.1b), tested, and then mounted on the back of a sports glove.
5. **Programming and pairing:** the transmitter and receiver programs were uploaded and the two HC-05 modules were paired.
6. **Testing and tuning:** the complete system was tested on a paved surface, and small adjustments were made after final assembly.



(a) Electronics bay under the seat



(b) Hand-unit circuit on a breadboard

Figure 7.1. Assembly of the receiver and transmitter electronics.

8 Cost Analysis

Table 8.1 gives the estimated component budget for the project. The PVC frame materials are not included in this estimate.

Table 8.1. Bill of materials and estimated cost.

No.	Component	Qty	Unit price (Tk)	Cost (Tk)
1	BO motor	2	85	170
2	Caster wheel	2	75	150
3	Battery holder	1	50	50
4	L298N motor driver	1	180	180
5	Arduino Uno R3	1	785	785
6	Bluetooth module HC-05	2	330	660
7	Jumper wires (set)	1	100	100
8	Breadboard	1	45	45
9	Arduino Nano	1	400	400
10	9 V battery	1	75	75
11	Accelerometer–gyroscope MPU-6050	1	200	200
			Total estimated cost	**2,815**

The two microcontroller boards and the pair of Bluetooth modules make up about 65% of the budget. The total cost is very low compared with commercial powered wheelchairs, which shows that gesture control can be added at small extra cost.

9 Results and Discussion

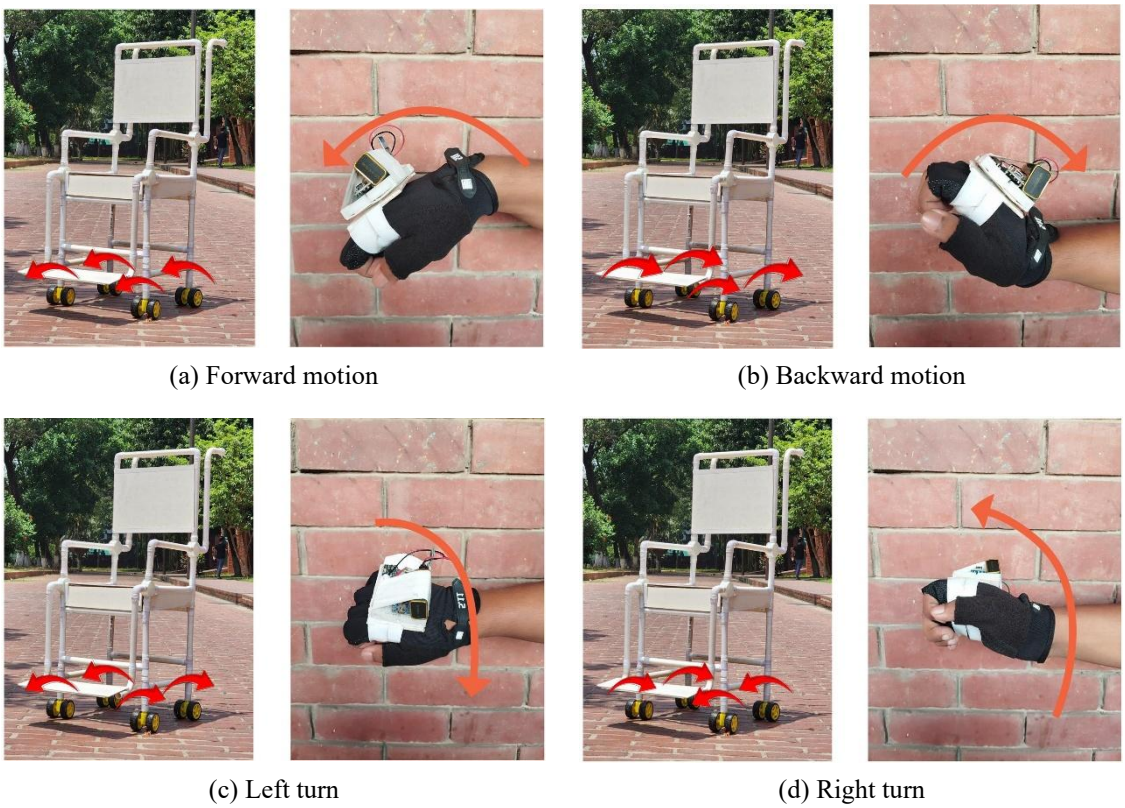
9.1 Test Results

The completed prototype (Figure 9.1) was tested on a paved brick surface. The response to each of the four gestures is shown in Figure 9.2. The main observations were:

1. The gesture system interpreted the hand gestures accurately, so the wheelchair movements could be controlled with precision.
2. The chair moved smoothly on the surface in response to the commands given by gesture.
3. The straight-line and turning speeds were kept constant at a manageable level.
4. Some small tuning of the chair was needed after final assembly.



Figure 9.1. Completed prototype of the gesture controlled wheelchair.



(a) Forward motion (b) Backward motion

(c) Left turn (d) Right turn

Figure 9.2. Wheelchair response (left of each pair) to the corresponding hand gesture (right of each pair).

9.2 Discussion

The results confirm that a wearable inertial sensor and a Bluetooth link can replace a joystick for basic wheelchair manoeuvres. Because the commands depend on tilt rather than on fine finger movement, the interface suits users who have limited grip strength. Using a neutral (level) hand position for stop makes the system fail safe: if the user relaxes the hand, the chair stops.

The current prototype has the following limitations:

- it is a proof-of-concept model with a PVC frame and small BO gear motors, so it is not designed to carry a human occupant;
- it has only one fixed speed, so it cannot slow down for precise manoeuvring or speed up on open ground;
- it has no obstacle detection, so the user is fully responsible for avoiding collisions; and
- the range of the Bluetooth link and the battery life limit the operating time and distance.

10 Conclusion and Future Scope

10.1 Conclusion

A gesture controlled wheelchair was designed, fabricated and tested. A glove fitted with an MPU-6050 sensor, an Arduino Nano and an HC-05 module detects hand tilt and transmits commands wirelessly to an Arduino Uno and L298N driver on the chair. The prototype moved forward, backward, left and right smoothly and accurately in response to the user's gestures, for an estimated component cost of about Tk 2,815. The project shows that intuitive, low-cost gesture control can improve the independence, convenience and quality of life of wheelchair users.

10.2 Future Scope

The system can be developed further in the following ways [2]:

- a speed controller can be added to provide different speed modes;
- a camera can be added for fully autonomous navigation;
- a laser scanner can be added to detect obstacles and avoid collisions; and
- a robotic arm can be installed on the wheelchair to give the user more assistance.

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